LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Brief description, purpose of a typical reservoir, EDP, hand pump, NRV pressure relief valve, automatic cut out valve, thermal relief valve
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Brief description, purpose of a typical reservoir, EDP, hand pump, NRV pressure relief valve, automatic cut out valve, thermal relief valve, sequence valve, throttling valve, accumulator jack selector valve and filters
5. Specific Objectives	17.1.2.1 Trainee will be able to describe typical hydraulic system and construction of its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of hydraulic components on schematic; Safety: Do not operate live systems; Supervision: Instructor monitors schematic tracing
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a hydraulic system in an aircraft?
(b) Name three types of valves used in hydraulic circuits.
(c) Why is fluid cleanliness critical in hydraulic systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Reservoir purpose and construction • EDP operation and construction • Hand pump mechanism and use	L & Ds	PPT, IB	15 min
(b)	• NRV and automatic cut out valve functions • Thermal relief valve operation • Sequence and throttling valve roles	L & Ds	PPT, PDF	20 min
(c)	• Accumulator jack selector valve function • Filter types and maintenance • System component integration overview	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between an NRV and a thermal relief valve.
(b) Describe the role of the accumulator jack selector valve in the system.
(c) How does a hand pump assist the EDP during system start-up?

12. Home Assignment and Follow Up	Prepare a brief note on the basic design requirements of the alighting gear to be discussed in LP 10.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Alighting gear, basic design requirements, types of undercarriage and functions	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min